

controlFK.m Documentation

controlFK.m

Purpose

Computes the robot TCP pose from the current joint angles.

$$q \rightarrow {}^B T_{TCP}$$

Inputs

- `robot` — robot DH parameters and TCP transform
- `q` — joint angles, 6×1

Output

- `T_B_TCP` — 4×4 homogeneous transformation from Base to TCP

Main Steps

1. Read:

- `a`
- `d`
- `alpha`
- `thetaOffset`

2. Start from:

```
T = eye(4);
```

1. For each joint:

```
theta = q(i) + thetaOffset(i);  
  
A = dhTransform(a(i), d(i), alpha(i), theta);
```

```
T = T * A;
```

1. Apply the Flange-to-TCP transform:

```
T_B_TCP = T * robot.tool.T_F_TCP;
```

Main Equation

$${}^B T_{TCP} = A_1 A_2 A_3 A_4 A_5 A_6 {}^F T_{TCP}$$

Notes

- Uses Standard DH.
- Does not use MATLAB `getTransform()` internally.
- MATLAB `rigidBodyTree` is only used for validation.
- Validated against:
 - MATLAB reference FK
 - `symbolicFK.m`

Related Files

- `dhTransform.m`
- `symbolicFK.m`
- `controlJacobian.m`